TEXORA · LECTURE SHEET

Jute Spinning: Drawing, Flyer Frames and Geotextiles

The long-staple counterpart of the cotton system: jute arrives as reeds metres long, so its machines are built around pinned faller bars that grip and carry the fibres — through drawing frames to flyer spinning frames — and its coarsest yarns now leave the mill as woven geotextiles that protect soil and river banks.

TE-2201 · SPINNING TECHNOLOGY-II · TOPIC 7 · LEVEL 2 TERM 2

1. Why jute needs its own machinery

Jute fibre reaches the mill as **reeds** — whole plant stems' worth of bast fibre, 1.5–4 m long — which batching, softening and carding (breaker and teaser/finisher cards, preceded in modern mills by cutting or stretching the reeds) reduce to a sliver of entangled, very long fibre strands. No roller drafting system can control strands of this length and variability: rollers set far enough apart to avoid breaking the fibre cannot grip and meter it. Jute machinery therefore drafts with **faller bars** — bars carrying rows of pins (gills) that comb into the sliver, carry it forward at a controlled speed, and support it throughout the drafting zone. Every jute drawing and spinning machine in this sheet is a variation on that one idea.

2. Jute drawing frames — objects and functions

- To **attenuate** the card sliver progressively toward spinning weight (jute sliver and yarn are weighed in the direct system — see Section 5).
- To **double** several slivers into one at each passage, averaging their weight variation — the main evenness tool in a system whose fibre is inherently variable.
- To **straighten, parallelise and clean** the fibre strands: the pins of successive faller bars comb the strands, open entanglements and let short fibre and dirt fall out.
- To deliver a uniform sliver, coiled in cans, of the weight the next passage (or the spinning frame) is set to receive.

Three passages are usual: **first (breaker) drawing**, **second (inter) drawing** and **finisher drawing**. Each passage doubles (typically 2–4 ends up, sometimes more) and drafts; drafts per passage are moderate — of the order of 5–10 — because aggressive drafting of long, entangled strands produces slubs rather than evenness.

3. Detailed study: how a jute drawing frame works

The feed slivers pass through a feed roller pair into the drafting field, where a **conveyor of faller bars** takes over. Each faller bar is screwed along the frame by twin screws; its pins rise into the sliver at the back of the field, travel forward carrying the fibre at the retaining rollers' surface speed, and drop away at the front. Two speeds therefore govern the strand: the slow fallers **retain** the fibre mass while the **drawing rollers** at the front pull the leading ends through the pins at the delivery speed. Draft is set by the ratio of those speeds:

$$\text{Draft} = \text{delivery (drawing-roller) surface speed} \div \text{feed (retaining-roller / faller) surface speed}$$

The pinned field is the control: fibres whose leading ends have been seized by the drawing rollers are drawn forward through the pins, sliding past the fibres still held back — a combing attenuation that no pair of rollers could impose on metre-long strands. At delivery, the sliver is condensed and coiled into a can, or, at the finisher, wound to the form the spinning frame wants.

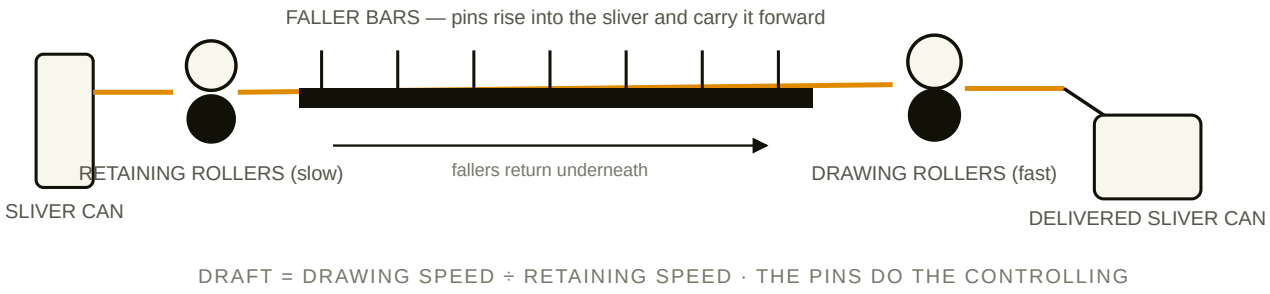


Figure 1 — A jute drawing frame's drafting field: slow retaining rollers feed the sliver into a moving bed of pinned faller bars; fast drawing rollers pull the leading fibres through the pins, producing the draft.

4. Drafting systems, drafts and roller settings

DRAFTING SYSTEM	HOW FIBRE IS CONTROLLED	WHERE USED
Screw-gill faller system	Full field of pinned faller bars between retaining and drawing rollers (Figure 1)	Drawing frames; the classic jute arrangement
Slip draft	Fibres are drawn through the faller field with partial slip permitted between fibre layers; simple and tolerant of variable material	Spinning frames for coarser yarns

Controlled (positive) draft	Retaining rollers plus aprons/extra pinned control hold the fibre until the draft point, minimising slip	Finer yarns, where evenness is worth the complexity
Apron draft	A pair of aprons replaces fallers in the final zone, giving smooth control of shorter fibre	Modern finisher and spinning frames for finer counts

Suitable drafts and settings. Because the "fibre length" is the cut reed length chosen in preparation (commonly worked in lengths around 8–16 inches after cutting), the rule mirrors cotton practice but at jute scale: the **reach** (distance between retaining nip and drawing nip) is set slightly longer than the longest fibre strands to be drafted, so fibres are pulled through the pins rather than broken between nips; set the reach shorter than the strand length and the machine literally tears fibre, raising short-fibre content and fly. Draft is built up over the three passages (each ≈ 5–10) rather than taken at once. Roller and faller settings, pin density (coarse pins for first drawing, finer at the finisher) and faller speed are all matched to the weight of sliver being carried: a heavy sliver in a fine-pinned finisher field chokes the pins and drafts in lumps.

5. Flyer spinning frames — objects, twisting, winding

The jute **spinning frame** is a flyer frame scaled for heavy strands. Its objects: to draft the finisher sliver to yarn weight (draft roughly 8–15, by the slip or controlled draft systems of Section 4), to **twist** the strand into yarn, and to wind it onto a bobbin for the next stage (weaving preparation or cop winding). The flyer — a large, robust version of the roving-frame flyer — rotates about the bobbin: each revolution inserts one turn of twist and, by the flyer's lead over the bobbin (or bobbin lead on some frames), winds the yarn on. **Bobbin building** follows the simplex principle adapted to coarse yarn: the bobbin rail traverses to lay parallel coils in layers and shortens its stroke progressively so the package ends taper; package sizes are large because jute yarn is consumed in bulk. Spindle speeds are far below cotton practice — the strands are heavy and the flyers big, so speeds of roughly 3,000–4,500 rpm are typical territory.

5.1 Amount of twist in jute yarns

Jute counts are quoted in the **direct system**: pounds per spindle of 14,400 yards — a higher number means a *coarser* yarn (the opposite of cotton's Ne). Twist is specified in turns per inch and is matched to end use: **weft yarns**, which meet little tension in weaving, carry about 2.4–3.0 tpi; **warp yarns**, which are abraded and tensioned throughout weaving, carry more, about 3.2–3.8 tpi; sacking yarns sit lower, hessian warp yarns in the middle of the warp range. Twist is a strength-versus-softness bargain in jute exactly as in cotton: too little and the coarse fibre strands slip apart; too much and the yarn turns hard and wiry, losing the soft, bulky handle sacking and hessian need.

6. Modern spinning systems for jute yarns — a brief outline

- **High-speed flyer frames** with automatic doffing, improved faller control and electronic draft setting — the traditional line, made faster and more consistent.
- **Apron-draft spinning frames** producing finer, evener jute yarns (down toward counts once thought impossible for jute), opening apparel and furnishing markets.
- **Ring spinning adaptations** for fine jute and jute-rich blends, borrowing cotton-system control for the finest work, alongside **blending lines** that spin jute with cotton, viscose or synthetics to combine jute's price and ecology with the partner fibre's softness.
- **Precision winding and electronic clearing** after spinning, bringing jute packages up to the delivery standards other fibres enjoy.

7. Geotextiles

A **geotextile** is a permeable textile used in contact with soil in civil engineering — to separate layers, filter water, drain, reinforce or protect. Jute geotextiles (JGT) exploit jute's virtues: cheap, locally abundant in Bangladesh, strong when new — and **biodegradable**, rotting away in one to a few seasons after vegetation has taken over the protection duty. This designed disappearance is their advantage over permanent synthetics where only temporary protection is needed.

TYPE	CONSTRUCTION	MAIN APPLICATIONS
Woven jute geotextile (soil saver)	Open-mesh woven fabric of coarse jute yarns, commonly about 500–1,200 g/m²	Erosion control on slopes and embankments; holds soil and moisture while grass and plants root through the mesh
Treated woven JGT	Woven jute, rot-proofed (e.g. bitumen or copper-based treatments) to extend field life	River bank and road-edge protection where a longer life is specified
Jute nonwoven / felt	Needle-punched fibre mat, often latex-backed	Mulch mats, plant protection, drainage and filtration layers, temporary road underlay

Typical deployments: stabilising road and railway embankments, protecting river banks against erosion until vegetation establishes, controlling surface erosion on construction sites and mine spoil, and as a separation layer in rural road construction trials. The engineer's specification states fabric weight, mesh opening and tensile strength; the environmental argument — a natural fibre replacing plastic in the soil — increasingly writes itself into tenders.

Glossary

Reed — the whole jute fibre strand as stripped from the plant stem, 1.5–4 m long. **Faller bar** — a bar carrying pins (gills) that supports and controls fibre through the drafting field. **Reach** — the distance between retaining and drawing nips in a jute drafting zone. **Slip draft / controlled draft** — drafting with partial fibre slip permitted / with positive fibre control to the draft point. **Spindle (jute count)** — a fixed length of 14,400 yd; jute count = weight in pounds of one spindle (direct system). **Hessian** — plain-woven jute fabric (lighter); **sacking** — heavier twill-woven jute fabric for bags. **Geotextile** — a permeable textile used with soil for separation, filtration, drainage, reinforcement or protection.

Key points

• Jute's metre-long fibres force pinned faller-bar drafting: fallers carry and control what rollers cannot grip.
• Drawing frames (three passages) double and draft moderately (≈ 5–10 per passage), comb the strands straight, and set sliver weight for spinning.
• Reach must exceed the longest strand; draft is the ratio of drawing to retaining surface speeds.
• Flyer frames spin jute as they do roving — draft, one twist per flyer revolution, tapered bobbin build — at far lower speeds and far heavier strands.
• Jute twist: weft ≈ 2.4–3.0 tpi, warp ≈ 3.2–3.8 tpi; count is direct (lb per 14,400 yd spindle), so higher count = coarser.
• Geotextiles are jute's modern growth market: woven soil savers and nonwovens give temporary, biodegradable erosion protection.

Practice

1. Explain why setting the reach shorter than the longest fibre strands damages the material, and what symptoms appear in the sliver.
2. A drawing frame's retaining rollers run at 2.0 m/min and its drawing rollers at 16 m/min. What is the draft? If three such passages each applied a draft of 8 with no doubling anywhere, what total draft would act between the card sliver and the finisher sliver — and why does real practice never do this?
3. Why does jute use three drawing passages of moderate draft rather than one passage of very high draft?
4. A buyer specifies a warp yarn and a weft yarn of the same jute count. Which gets the higher twist, by roughly how much, and why?
5. A district engineer needs slope protection for one monsoon season only, after which grass cover is planned. Recommend a product from this sheet and justify it against a synthetic geotextile.

Answer key

1. With the reach shorter than the strand, a fibre can be gripped by the drawing rollers while its tail is still firmly held by the retaining nip and the pin field; the draft then tears the strand instead of sliding it. Symptoms: increased short-fibre content, fly and dust in the frame, slubs and weight variation in the sliver, and higher waste.
2. Draft = 16 ÷ 2.0 = **8**. Without any doubling, three passages of 8 would multiply: 8 × 8 × 8 = **512** total draft — in reality doublings at each passage feed several slivers in, restoring the weight that each draft removes; draft and doubling are always set together.
3. Long, entangled jute strands cannot be attenuated drastically in one step: the pins would have to hold an enormous length difference, fibres would tear, and the draft would emerge as lumps and slubs. Moderate successive drafts, with doubling between them to re-average the weight, straighten and open the strands gradually — each passage leaves the material in a state the next passage can draft further.
4. The warp yarn — warp ≈ 3.2–3.8 tpi against weft ≈ 2.4–3.0 tpi, roughly 0.8 tpi more. Warp yarn is held under sustained tension and abraded by heddles, reed and shuttle action throughout weaving, so it needs the extra cohesion and abrasion resistance that twist gives; weft meets little stress and is kept softer.
5. A woven jute soil saver of suitable weight (≈ 500–1,200 g/m²): it holds the soil surface, conserves moisture and lets the grass root through its open mesh during the season; when the vegetation takes over, the jute biodegrades in place instead of leaving plastic in the soil to be retrieved — at lower material cost, using a local fibre.
